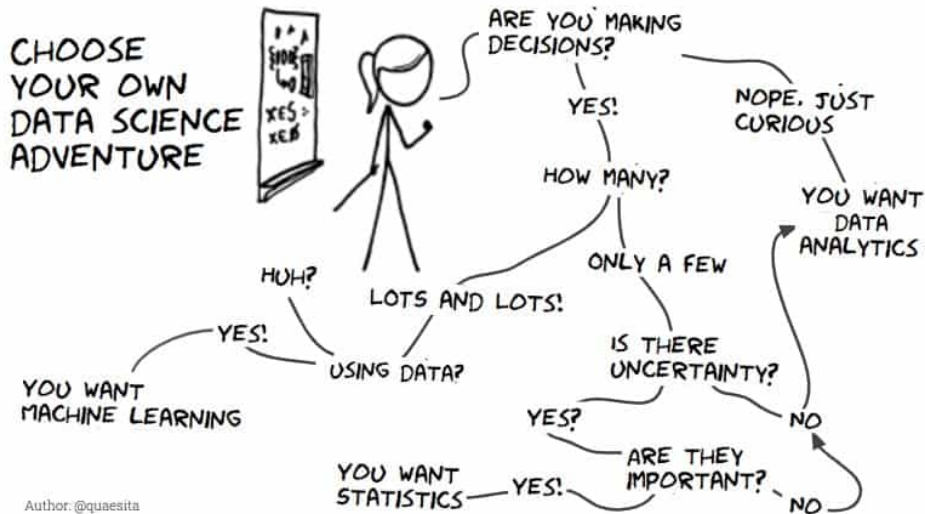


BIOS 835: Linear Classification Methods

Alexander McLain

September 16, 2025



CLASSIFICATION

Classification Introduction

- ▶ We'll now switch from continuous to binary (yes/no) types outcomes.
- ▶ For example, the Wisconsin Breast Cancer Dataset (WBCD) was created by Dr. William H. Wolberg at the University of Wisconsin Hospitals, Madison.
- ▶ These data are used to predict whether a breast cancer tumor is benign or malignant based on various cell features.

Classification Introduction

- ▶ We'll now switch from continuous to binary (yes/no) types outcomes.
- ▶ For example, the Wisconsin Breast Cancer Dataset (WBCD) was created by Dr. William H. Wolberg at the University of Wisconsin Hospitals, Madison.
- ▶ These data are used to predict whether a breast cancer tumor is benign or malignant based on various cell features.
- ▶ **Examples of cell features:**
 - ▶ Radius (mean of distances from the center to points on the perimeter),
 - ▶ Texture (standard deviation of gray-scale values),
 - ▶ Perimeter, Area, etc.

Each of these features is calculated for three different metrics: mean, standard error, and the “worst” or largest value.

Classification Introduction

- ▶ Classification is the process of predicting the class label of a given input based on training data that contains input-output pairs.
- ▶ It involves building a model that learns from the training data and can predict the class of new, unseen data.
- ▶ Linear regression cannot be used for classification.
- ▶ Some Common Algorithms: K-Nearest Neighbors (KNN), logistic regression, decision trees, Naive Bayes, Neural Networks, Gradient Boosting Machines

Logistic Regression

- ▶ If we are classifying using logistic regression, we have

$$\pi_i = \Pr(Y_i = 1 | \mathbf{x}_i) = \frac{\exp(\beta_0 + \beta'_1 \mathbf{x}_i)}{1 + \exp(\beta_0 + \beta'_1 \mathbf{x}_i)}$$

- ▶ The parameters are usually estimated using maximum likelihood methods, which give $\hat{\pi}_i$.

Model Selection in Logistic Regression

- ▶ The same questions about model selection come up.
- ▶ For example, which of the cell features should be used for our $\mathbf{x}$? Which summary measure should we use?
- ▶ The L_1 Lasso penalty discussed previously can be used for variable selection and shrinkage for the logistic regression model.

Model Selection Criteria

1. AIC/BIC

- ▶ Akaike information criterion (AIC) and Bayesian information criterion (BIC) judge a model by the value of the negative log-likelihood ($-\log\{L(\beta)\} > 0$).
- ▶ The formulas are

$$AIC = -2 \log\{L(\hat{\beta})\} + 2p$$

$$BIC = -2 \log\{L(\hat{\beta})\} + p \log(n)$$

- ▶ BIC penalizes more heavily for adding additional parameters.
- ▶ We choose the model with the lowest AIC/BIC.
- ▶ These can be used with forward/backward/step-wise selection procedures.

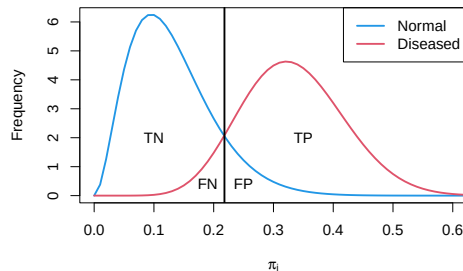
Measuring of predictive ability

- ▶ We will now take a bit of a tangent to discuss quantifying the predictive ability of a model.
- ▶ Some of these methods will take more background than others.
- ▶ In general, they are all trying to answer the following:

Ten new subjects walk into the room (all were not in the data used to estimate $\hat{\beta}$), which model is going to give me the best estimate of the predictive probability (i.e., $\hat{\pi}_i$) for those ten subjects?

Sensitivity and specificity

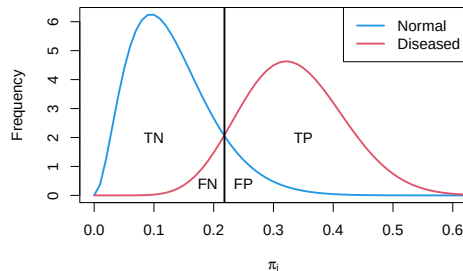
- ▶ Imagine a study evaluating a new test that screens people for a disease $\{0, 1\}$.
- ▶ The test outcome can be positive (predicting that the person has the disease) or negative (predicting that the person does not have the disease).
- ▶ The test results for each subject may or may not match the subject's actual status.



Sensitivity and specificity

In this setting, there are 4 things that could happen:

- ▶ **True positive (TP):** Sick people correctly diagnosed as sick
- ▶ **False positive (FP):** Healthy people incorrectly identified as sick
- ▶ **True negative (TN):** Healthy people correctly identified as healthy
- ▶ **False negative (FN):** Sick people incorrectly identified as healthy



2x2 Classification

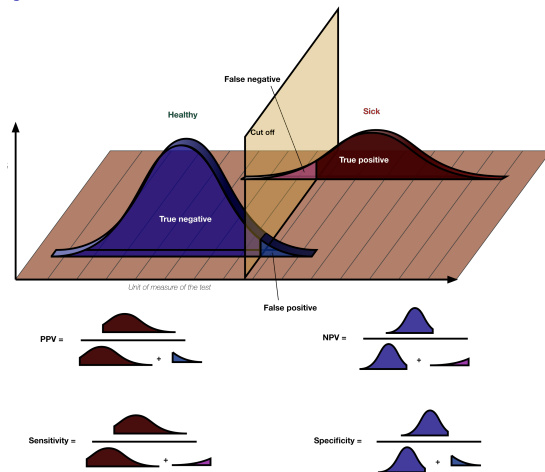
	Test Pos. (T+)	Test Neg. (T-)	Totals
Diseased (D+)	TP	FN	TP+FN
Not diseased (D-)	FP	TN	FP+TN
Totals	TP+FP	FN+TN	n

Where:

- ▶ TP+FN: **number of sick people.**
- ▶ FP+TN: **number of health people.**
- ▶ TP+FP: number that tested positive.
- ▶ TN+FN: number that tested negative.

We'll use FP, FN, FP and TN to quantify the predictive value model.

Sensitivity, specificity, PPV, NPV



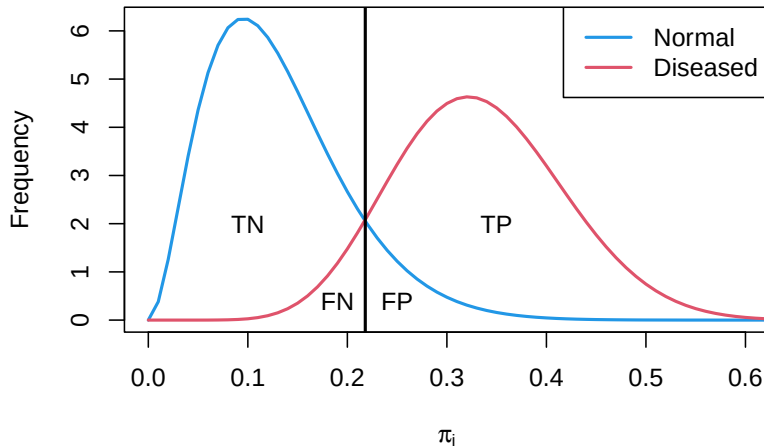
Estimating sensitivity and specificity from a logistic model

- ▶ To estimate sensitivity and specificity from a logistic model we we'll use the values of $\hat{\pi}_i$ as the results of the “test.”
- ▶ Let's assume that $Y = 1$ if the person has the disease, so that a high value of $\hat{\pi}_i$ is indicative that the person will have the disease.
- ▶ Then we can pick a value, say π_0 , such that the result of the test is

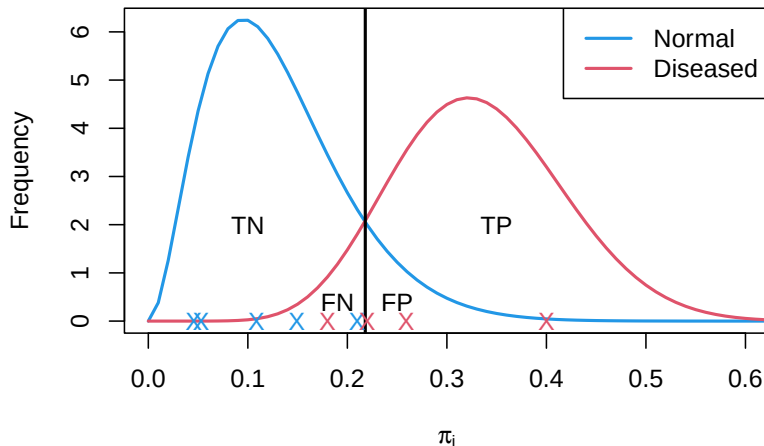
$$T_i = \begin{cases} +, & \text{if } \hat{\pi}_i > \pi_0 \\ -, & \text{if } \hat{\pi}_i \leq \pi_0 \end{cases}$$

- ▶ So all people with a predicted value of at least π_0 will have a test equal to “positive”.

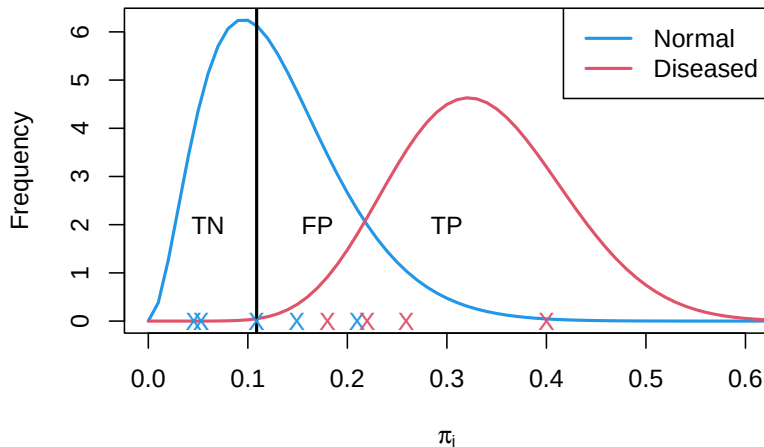
Classification of a test



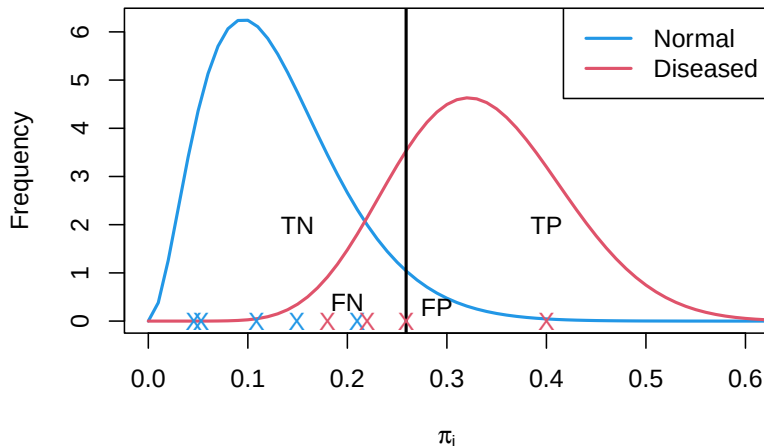
Classification of a test



Classification of a test



Classification of a test



Confusion Matrix and π_0

For each value of π_0

$$T_i = \begin{cases} +, & \text{if } \hat{\pi}_i > \pi_0 \\ -, & \text{if } \hat{\pi}_i \leq \pi_0 \end{cases}$$

we get a different confusion matrix:

	T+	T-	Totals
D+	TP(π_0)	FN(π_0)	# Diseased
D-	FP(π_0)	TN(π_0)	# Not Diseased
Totals	# Pos tests(π_0)	# Neg tests(π_0)	n

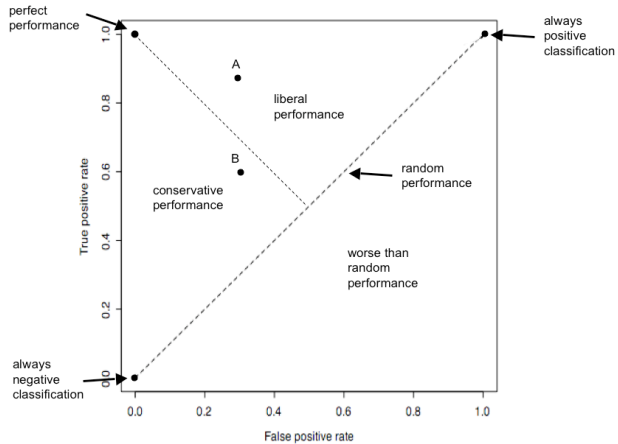
- ▶ The number of negative tests increases as π_0 increases.
- ▶ The number of positive tests decreases as π_0 increases.
- ▶ When $\pi_0 = 0$, the # Pos tests(0) = $n \rightarrow TN(0) = 0$
- ▶ When $\pi_0 = 1$, the # Neg tests(1) = $n \rightarrow TP(1) = 0$

Receiver Operating Characteristic (ROC) Curves

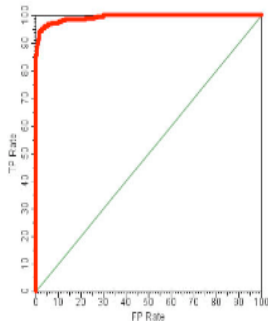
- ▶ Receiver Operating Characteristic (ROC) Curves are a way to measure the predictive ability of a model.
- ▶ ROC curves look at the $Sens(\pi_0)$ and $1 - Spec(\pi_0)$ for all values of π_0 .
- ▶ When $\pi_0 = 0$ all tests are positive, $Sens(0) = 1$ and $1 - Spec(1) = 1$.
- ▶ When $\pi_0 = 1$ all tests are negative, $Sens(1) = 0$ and $1 - Spec(1) = 0$.

Motivation for ROC Curves

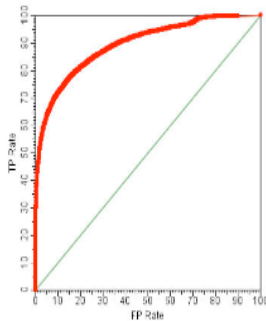
- ▶ For example, assume we have 100 diseased and 100 not diseased in our sample.
- ▶ Suppose that we list all 200 people from smallest to largest $\hat{\pi}_i$.
- ▶ Suppose that when I increase π_0 from 0.4 to 0.5 an additional 20 people have T^- .
- ▶ If the test is worthless I would expect 10 of those people to be diseased and 10 not diseased.
- ▶ In general, (If the test is worthless) as π_0 increases, 50% of the people are diseased and 50% are not diseased.



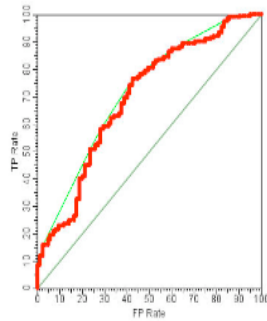
ROC curve examples



Almost perfect
prediction



Good prediction



Poor prediction

Quantifying ROC curve performance

- ▶ Notice that the models that predict better have larger area under the ROC curve (AUC).
- ▶ AUC is the main way to quantify an ROC curve.
- ▶ The AUC is equal to the probability that the model will rank a randomly chosen diseased individual higher than a randomly chosen healthy individual.

$$\Pr(\hat{\pi}_i > \hat{\pi}_j | Y_i = 1 \text{ and } Y_j = 0)$$

AUC values

- ▶ We will interpret the AUC in terms of the discriminative ability of the model.
- ▶ That is, how well does the model discriminate between diseased and non-diseased individuals?

$\widehat{AUC}$	Interpretation of discriminative ability
< 0.5	worse than expected by chance
$= 0.5$	no discrimination
$0.7 - 0.8$	acceptable discrimination
$0.8 - 0.9$	excellent discrimination
> 0.9	outstanding discrimination

Cross-Validation for Logistic Regression Models

- ▶ Suppose we have several candidate logistic regression models (different sets of predictors, different functional forms).
- ▶ To compare them fairly, we need to evaluate performance on data not used to fit the model.
- ▶ **K-fold Cross-Validation:**
 - ▶ Split the data into K folds.
 - ▶ Each fold is the validation set once, with the others used for training.
 - ▶ Average the results across folds.
- ▶ For each model, we can compute performance metrics in each fold:
 - ▶ AUC
 - ▶ Sensitivity, Specificity
 - ▶ Positive Predictive Value (PPV), Negative Predictive Value (NPV)

Using Metrics to Select the Best Model

- ▶ After cross-validation, each candidate model has averaged performance metrics.
- ▶ **Choose the model** based on:
 - ▶ Highest AUC (if overall discrimination is the priority).
 - ▶ Best sensitivity/specificity balance (if identifying positives/negatives is critical).
 - ▶ Highest PPV/NPV (if predictive value matters most in the public health context).
- ▶ Key point: the “best” model depends on which metric aligns with the study goal.
- ▶ Example:
 - ▶ For screening, prioritize sensitivity/NPV.
 - ▶ For confirming disease, prioritize specificity/PPV.
- ▶ Cross-validation ensures these metrics reflect how the model will perform on new data.

MLE Logistic Regression

- ▶ In logistic regression, β can be estimated by maximizing the log-likelihood

$$\log\{L(\beta)\} = \sum_{i=1}^N \log \{ \pi_i^{y_i} (1 - \pi_i)^{1-y_i} \}$$

- ▶ This can be written in terms of $\beta = \{\beta_{01}, \beta_1\}$ as

$$\log\{L(\beta)\} = \sum_{i=1}^N \left\{ y_i \beta' \mathbf{x}_i - \log(1 + e^{\beta' \mathbf{x}_i}) \right\}$$

where $\mathbf{x}$ now contains an intercept term.

- ▶ β is usually estimated using an algorithm called *iteratively reweighted least squares* or IRLS.

L_1 penalized logistic regression

- ▶ The L_1 Lasso penalty discussed previously can be used for variable selection and shrinkage.
- ▶ For the 2-class case, we look to find

$$\max_{\boldsymbol{\beta}} \left[\sum_{i=1}^N \left\{ y_i \boldsymbol{\beta}' \mathbf{x}_i - \log(1 + e^{\boldsymbol{\beta}' \mathbf{x}_i}) \right\} - \lambda \sum_{j=1}^p |\beta_j| \right]$$

- ▶ This is a difficult concave optimization, but the IRLS solution can be applied.
- ▶ This leads to a repeated application of a weighted lasso algorithm.

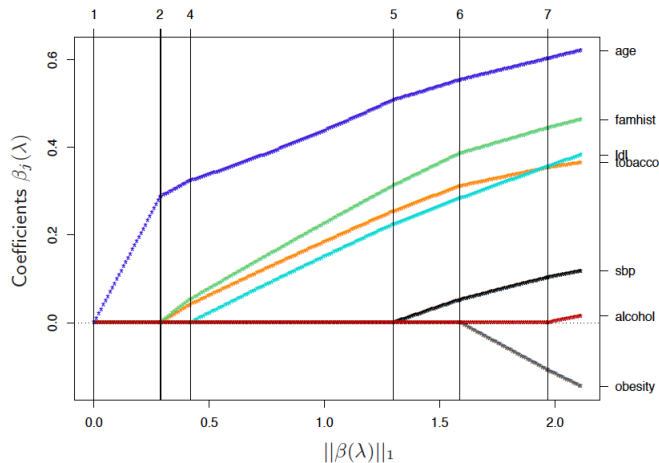


Figure: From ESL (online version). Note that the paths are not linear.

Linear Discriminant Analysis (LDA): Setup

- ▶ Suppose we have data from K groups (e.g., smokers vs. nonsmokers).
- ▶ We model predictors X as coming from a multivariate normal (MVN) distribution for each group:

$$X \sim MVN(\mu_k, \Sigma_k), \quad k = 1, \dots, K.$$

- ▶ If all groups share the same covariance matrix ($\Sigma_k = \Sigma$), then the boundaries between groups turn out to be **linear**.
- ▶ This is the key idea of **Linear Discriminant Analysis (LDA)**.

Linear Discriminant Function

- ▶ LDA builds a **discriminant function** for each group:

$$\delta_k(x) = x' \Sigma^{-1} \mu_k - \frac{1}{2} \mu_k' \Sigma^{-1} \mu_k + \log \pi_k$$

where

- ▶ μ_k = mean vector for group k ,
 - ▶ π_k = prior probability of being in group k ,
 - ▶ Σ = common covariance matrix.
- ▶ The classification rule:

Assign x to the group with the largest $\delta_k(x)$.

- ▶ In practice: this means each group is separated by a **linear boundary**.

Linear Discriminant Analysis Example

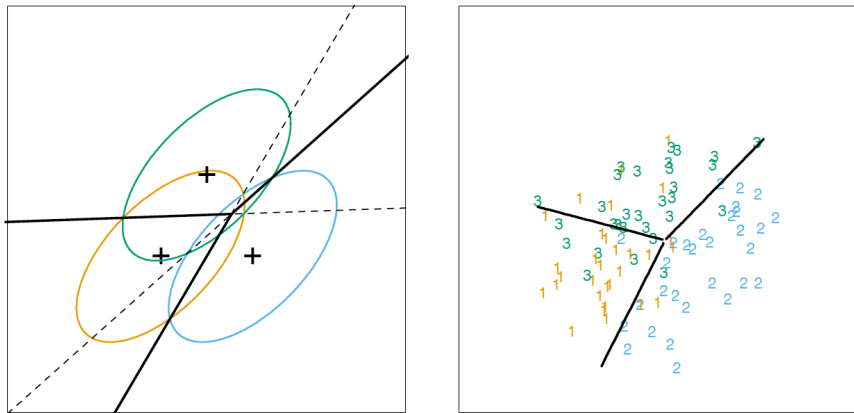


Figure: From ESL (online version).

Estimating the Parameters in LDA

- ▶ Estimate group priors: $\hat{\pi}_k = N_k/N$.
- ▶ Estimate group means: $\hat{\mu}_k = \text{average of } x_i \text{ in group } k$.
- ▶ Estimate pooled covariance: $\hat{\Sigma} = \text{weighted average of within-group covariance}$.
- ▶ Note:
 - ▶ LDA does not rely heavily on the normality assumption for prediction.
 - ▶ The “cutpoint” (intercept) is more sensitive to distributional assumptions — often chosen or checked with cross-validation.

Quadratic Discriminant Analysis (QDA)

- ▶ If we allow each group to have its own covariance matrix (Σ_k), then the discriminant function becomes **quadratic**.
- ▶ The classification rule is still to assign x to the group with the largest $\delta_k(x)$, but the boundary curves can now be **non-linear**.
- ▶ Compared to LDA:
 - ▶ More flexible, captures complex separation between groups.
 - ▶ Requires estimating more parameters, so can overfit with small data.

LDA vs QDA: A Tradeoff

- ▶ **LDA**: simpler, linear boundaries, fewer parameters. Works well when groups are roughly separated by lines or planes.
- ▶ **QDA**: more flexible, quadratic boundaries, more parameters. Works well when groups need curved boundaries to be separated.
- ▶ Which to choose?
 - ▶ Try both and compare using cross-validation.
 - ▶ LDA is often more stable when sample size is small.